Offshore Radar Anomaly Surveillance Reporting Improvement

INTRODUCTION

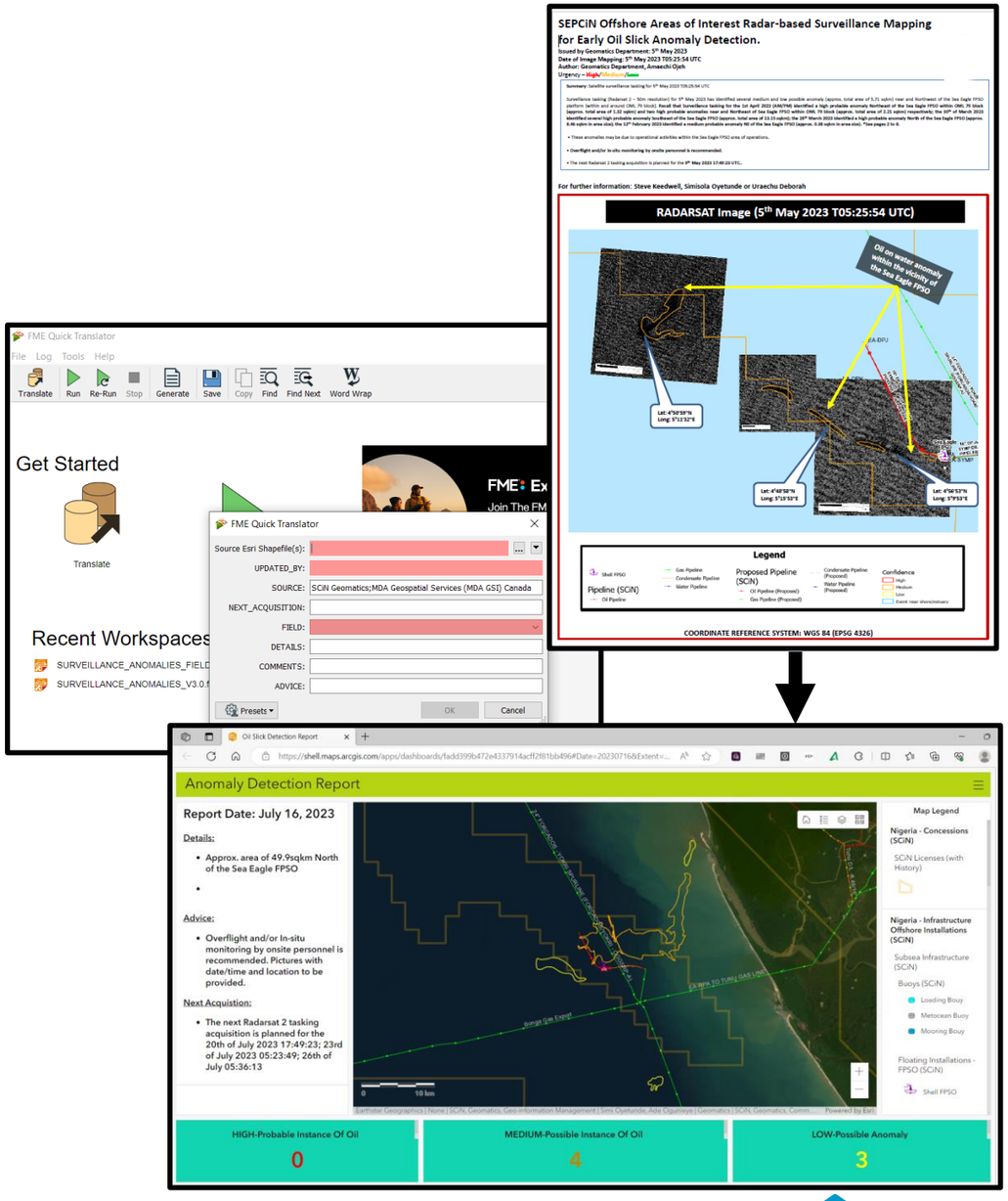
- Offshore radar surveillance provides detection of oil slick and seep anomalies potentially associated with SCiN offshore assets and operations. The report/alert delivered to stakeholders facilitates a rapid response and investigation of any potential integrity issue thereby safeguarding the environment and the company's reputation. The monitoring program also provides detection of surface oil from third-party activity and incidents.
- Geomatics provides support to spatialize and analyze the data, leveraging overlays with other spatial datasets on an interactive dashboard. To further enhance the way the reports are delivered for improvement in efficiency, turn-around time and for better visualization of the spatial datasets, the GIM team developed an improved workflow to convey an instance of the anomaly detection report that supports stakeholders' decision and interpretation processes.

BUSINESS CHALLENGE

- Opportunity to enhance anomaly data processing, report analysis and visualization of the Radar anomaly observations, through:
  - Prompt turnaround in report presentation and delivery.
  - User-friendly interactive display and addition of other relevant spatial datasets to show spatial interaction and proximity.
  - Replacement of pdf reports with a more interactive portal for effective reporting, monitoring incidents, decision making and other analysis.

BUSINESS SOLUTION

- Timely processing (through FME automation), interpretation and reporting of significant suspected crude anomaly, and integration with other ancillary data (SCiN and Third-Party) for accurate (>80%) determination of the likely source and information/reports accessible to key stakeholders.
- Controlled, user-friendly and interactive interface (ArcGIS Dashboard) for interacting with and visualizing various spatial data and anomaly statistics required to be informed or make decisions.
- Enhanced portal with advance widgets for reporting, monitoring incidents, decision making and other analysis.

BUSINESS VALUE

- Goal Zero: No harm, less exposure.
- Demonstrate Shell's stewardship of the environment to government regulators and other operators.
- Maintain record of offshore environment in SCiN Corporate GIS to replay in the event of future litigation.
- Possible cost saving in improving near real-time situation awareness, optimizing resource allocation and will enhance operational efficiency reduce response time and lead to cost saving in surveillance operation.
- Improved processing and delivery time from >2hrs to ~ 30minutes after acquisition.